Rupali Patel

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Education

VIT Bhopal University — **Bhopal, Madhya Pradesh**
B.Tech Electronics and Communication Engineering — Expected June 2027
Cumulative GPA: 9.23/10

Technical Skills and Tools

Language: C++, Java
Tools: LTspice, Proteus
Coursework: Embedded System, Iot, Signal Processing

Projects

Pulse Oximeter — **July 2024- Sep 2024**

- Description: Designed and implemented a pulse oximeter system using Arduino with integrated sensor interfacing to measure heart rate and blood oxygen (SpO_2) levels in real time.
- Developed an embedded system to acquire physiological signals from the MAX30100 sensor and process them using Arduino.
- Implemented real-time display of heart rate and SpO_2 values on an LCD module for easy user interaction.
- Ensured reliable I2C communication between Arduino and MAX30100 sensor module.
- Performed testing and validation of sensor readings to improve accuracy and stability of measurements.
- Technology: Arduino, MAX30100, LCD, Embedded C, I2C Protocol, Basic Electronic Components.

Object Detection In Water — **Sep 2024- Dec 2024**

- Description**:** Developed an Arduino-based underwater object detection system to sense and identify submerged objects using connected sensors and control electronics.
- Designed and implemented a real-time detection mechanism to measure distance and presence of objects in water environments.
- Technology: Arduino IDE, Ultrasonic Sensor, Embedded C, Java (for monitoring/interface), Serial Communication, Basic Electronic Components.
- Role : Hardware Designer – designed the complete circuit, selected components, and handled sensor interfacing with Arduino.

Sign Language to Speech Conversion System — **Jan 2025-Apr 2025**

- Description: Created a system that translates sign language into text and speech, enabling effective communication between deaf/mute individuals and non-sign language users.
- Designed a sensor-based hand gesture input mechanism using flex sensors to capture finger movements.
- Enabled wireless data transmission of recognized gestures using the HC-05 Bluetooth module.
- Technology: Arduino IDE, C++, Flex Sensors, HC-05 Bluetooth Module, Serial Communication, Basic Electronic Components.
- Role: Hardware integration of flex sensors and Bluetooth module with the microcontroller.
- Testing, debugging, and troubleshooting of hardware and communication issues.

Experience

Embedded System Design Internship-Maven Silicon — **Jan 2025-March 2025**

- Developed embedded firmware and hardware interfacing for **Alert system for women's** safety project.
- Improved skill level in microcontroller-based system design and embedded programming.
- Link- https://drive.google.com/file/d/1m3kuM5QLx20ST_MdDpKuo79MoKr2HIyt/view?usp=sharing

Achievements

- Awarded with Dreamed STAR Scholarship from VIT Bhopal.
- Secured first position in district merit in class 12^{th} .
- Awarded with **Reliance Foundation Scholarship** for academic excellence.

Additional Information

Languages: Hindi, English
Hobbies: Gaming, Reading books, Sketching